



In addition to Part I (General Handout for all courses appended to the timetable) this handout gives further details regarding the course.

Course No : CS F422
Course Title : **Parallel Computing**
Instructor-in-charge : **Dr. Apurba Das**

1. Scope and Objectives of the Course:

Parallel computing architectures have emerged as alternative to high performance computing using powerful single processor machines. Sequential algorithms i.e., algorithms designed for a single processor machine, do not harness the full potential of a parallel machine and hence the need to devise new parallel algorithms. Parallel algorithms are highly architecture dependent. Moreover, for a given problem, some parallel architecture is better suited than others. Therefore, it is necessary to study parallel architectures and techniques for designing efficient parallel algorithms.

The main objectives of this course are to give the students exposure to

- Models of parallel computers; Interconnection networks, basic communication operations
- Introduction to parallel algorithms; Parallel programming paradigms; issues in implementing algorithms on parallel computers
- Parallel programming with message passing interface; Performance analysis
- Scalability analysis; Basic design techniques for parallel algorithms
- Parallel algorithms for selected applications like sorting, searching and merging, matrix algebra, graphs.

2. Text Book:

T1: “*Introduction to Parallel Computing*”, Ananth Grama, Anshul Gupta, George Karypis and Vipin Kumar, Second Edition, Pearson Education, 2011.

3. Reference Books:

- R1.** M.J. Quinn, “Parallel Computing: Theory & Practice”, McGraw Hill Inc. 2nd Edition, Reprint 2017.
R2. M.J. Quinn, “Parallel Programming in C with MPI & OPENMP”, Jaico Books, 2004. (Reprint 2017).
R3. Kai Hwang and Faye A Briggs, “Computer Architecture and Parallel Processing”, Tata Mc Graw Hill Edition, 2012.
R4. Peterson, “Introduction to Parallel Computing – A Practical Guide with Examples in C”, Oxford University Press, 2008.
R5. Peter S Pacheco, “An Introduction to Parallel Programming”, Morgan Kaufmann Publishers, 2018.
R6 Wen-mei W. Hwu, David B. Kirk, and Izzat El Hajj, “Programming Massively Parallel Processors”



4. Course Plan:

Lecture No.	Learning Objectives	Topics to be covered	Chapter in the Text Book
1	Introducing Parallel Algorithms, studying algorithms, minimizing number of processors, Multicore Architecture, GPU architectures, Unified Memory,	Introduction to parallel processing and parallel processing terminology	R1 Sec. 1.1,1.2 & 1.3, T1 Chapter 1
2,3		Contrast between Data Parallelism & Control Parallelism	R1 Sec. 1.3 & 1.4
4-6		Parallel Programming Platforms – Multicore, GPU, and hybrid platforms	T1 Sec. 2.1 – 2.3, Lecture Notes
7		Physical Organization of parallel computer	T1 Sec. 2.4, Lecture Notes
8,9		PRAM algorithms, (parallel reduction, prefix sums, list ranking etc.)	R1 Sec. 2.3
10		PRAM Sorting Algorithms	Lecture Notes
11-12		Routing in parallel computer	T1 Sec. 2.5-2.6
13-15	Studying different organizations, mappings between them, data decomposition and	Processor-Processor mapping & mapping techniques	T1 Sec. 2.7
16-18		Decomposition Techniques and Task Mapping	T1 Sec. 3.1,3.2, 3.3
19-20	Performance metrics for parallel systems	Performance metrics for parallel system	T1 Sec. 5.1, 5.2
21-22		Iso-efficiency function & scalability issues	T1 Sec. 5.3, 5.4
23-24		Other scalability metrics	T1 Sec. 5.5, 5.6
25-26	Studying & Analyzing parallel versions of standard sequential algorithms on different systems - Multicore and GPU	Simple parallel algorithms on mesh and hypercube	R1 Chapter 6 R2, R3
27-30		Parallel Matrix Algorithms: Matrix Multiplication, Solving System of Linear Equations	T1 Chapter 8 R4
31-35		Sorting Algorithms on mesh and hypercube, Multicore, and GPU	T1 Chapter 9
36-38		Parallel Graph Algorithms: Prim's MST algorithm (parallelization), Dijkstra's shortest path algorithm (parallelization) etc.	T1 Sec. 10.1–10.4 R5
39-40	Recent Advances in Parallel Computing	High performance parallel computing – Case Study	Recent research publications



5. Evaluation Scheme:

Component	Duration	Weightage (%)	Date & Time	Nature of Component
Mid Sem Test	1 hr 30 min	30%	10/10 - 2.00 - 3.30PM	Closed Book
Continuous Assessment (Class)		10%	TBA	Open Book
Term Project		20%	TBA	Open Book
Comprehensive Exam	3 hrs	40%	16/12 FN	Closed Book

Note: For the Assignments/Term Projects (Take Home), exposure to basic programming would be useful.

Note: At least 40% of the evaluation component will be considered for Midterm Evaluation.

6. Chamber Consultation Hour: Friday 12-1 pm.

7. Notices: Notices regarding the course will be put up on the LMS/Google Classroom

8. Makeup Policy: No makeup exam allowed without prior permission.

9. Academic Honesty and Integrity Policy: Academic honesty and integrity are to be maintained by all the students throughout the semester and no type of academic dishonesty is acceptable.

INSTRUCTOR-IN-CHARGE

